Math 521, Section 2
Homework 13

Going forward, you may assume (no proof necessary) the familiar differentiation properties of e^x, $\log x$, $\sin x$, and $\cos x$ from calculus.

Problem 1. Consider the sequence of functions $f_n : [0,1] \to \mathbb{R}$ given by

$$f_n(x) = \begin{cases} nx & \text{if } x \in [0, \frac{1}{n}], \\ 2 - nx & \text{if } x \in (\frac{1}{n}, \frac{2}{n}], \\ 0 & \text{if } x \in (\frac{2}{n}, 1]. \end{cases}$$

(a) Prove that f_n converges pointwise to the function $f : [0,1] \to \mathbb{R}$, $f(x) = 0$.
(b) Prove that f_n does not converge uniformly to any function $g : [0,1] \to \mathbb{R}$.

Problem 2. Consider the series of functions $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{1+x^n}$.

(a) Prove that for any $r \in (0,1)$, the series converges uniformly on $[0,r]$.
(b) Prove that the series does not converge uniformly on $[0,1)$.

Problem 3. Consider the power series $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$.

(a) Find its radius of convergence R.
(b) Determine if the series converges at each of the endpoints $x = R$ and $x = -R$.

Problem 4. For each of the following power series, determine the radius of convergence.

$$\text{(a)} \quad \sum_{n=0}^{\infty} n^2 x^n \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \left(\frac{x}{n}\right)^n \qquad \text{(c)} \quad \sum_{n=0}^{\infty} n!\, x^n$$

Problem 5. The *hyperbolic sine* function is defined by $\sinh x = \frac{1}{2}(e^x - e^{-x})$.

(a) Find its Taylor series centered at $x = 0$.
(b) Prove that this series converges to $\sinh x$ for any $x \in \mathbb{R}$.

Problem 6. Consider the function $f : (-1, \infty) \to \mathbb{R}$, $f(x) = \log(1+x)$.

(a) Using induction, prove that $f^{(n)}(x) = (-1)^{n+1}(n-1)!\,(1+x)^{-n}$ for all $n \geq 1$.
(b) Using part (a), find the Taylor series of f centered at $x = 0$.
(c) Suppose we want to approximate the number $f(1) = \log 2$. Prove that for any $n \geq 2$,

$$\left| f(1) - \sum_{k=1}^{n-1} \frac{(-1)^{k+1}}{k} \right| \leq \frac{1}{n}.$$

This gives us a way to approximate $\log 2$ and to guarantee that the approximation is as accurate as we want, without having to know the value of $\log 2$ beforehand.